Exercise 1:

Consider an object M that is moving on a horizontal straight trajectory. The position of M is located on the axis $\overrightarrow{x'x}$ taken along this trajectory.

At the instant $t_1 = 2s$, the object was at point M_1, then at the instant $t_2 = 6s$, the object becomes at M_2 as shown by the figure of doc.1.

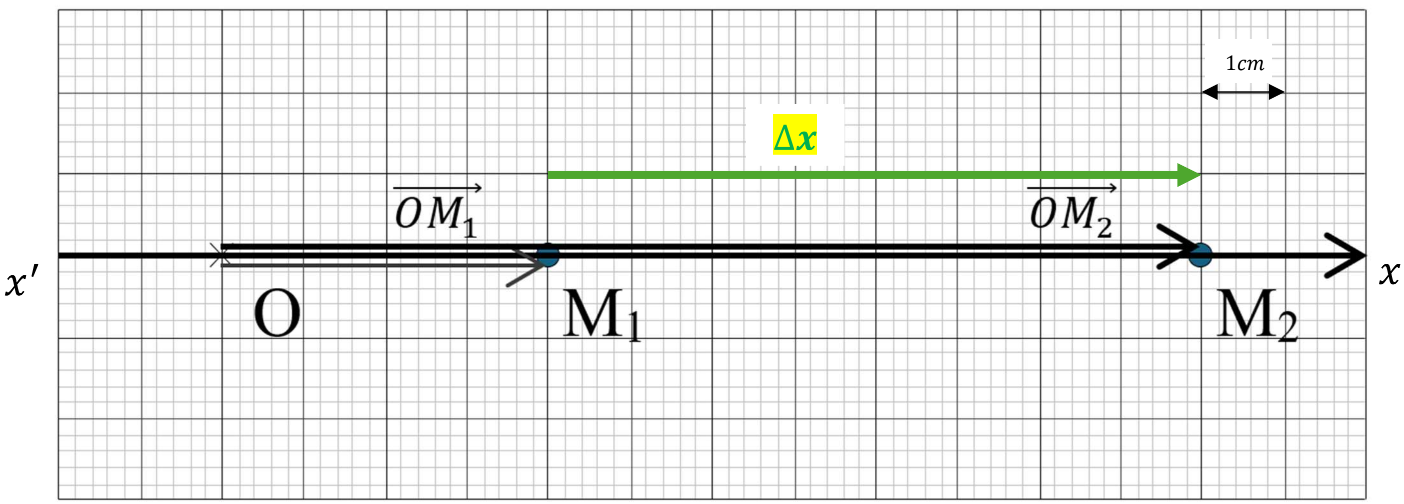
1- What is the position of the mobile at the instants t_1 and t_2.

$x_1 = \overline{OM}_1 = 4\ cm$

$x_2 = \overline{OM}_2 = 12\ cm$

2- Calculate the displacement between M_1 and M_2. Plot the corresponding displacement vector.

$\Delta x = x_2 - x_1 = 12 - 4 = +8\ cm$



Doc. 1

3- Calculate the value of the average speed between t_1 and t_2.

$$v_{av} = \frac{d}{\Delta t} = \frac{M_1M_2}{t_2 - t_1} \Rightarrow v_{av} = \frac{8\ cm}{6\ s - 2\ s} = \frac{8\ cm}{4\ s} = 2\ cm/s$$

Exercise 2:

The figure below reproduces, to real scale, the successive positions, separated by the same interval of time $\tau = 50ms$, of a mobile A on a horizontal table.

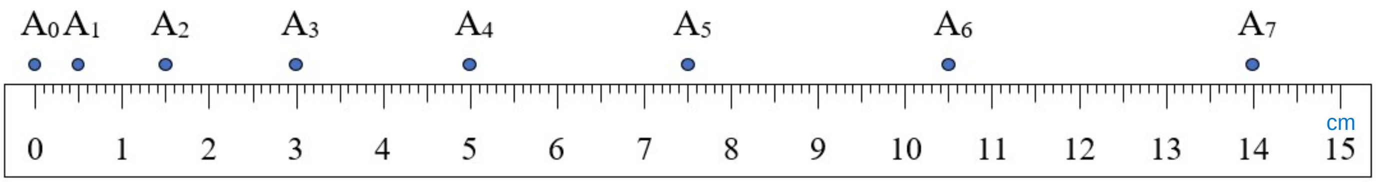
1. What is the shape of the trajectory?

   Rectilinear, since the successive positions occupied by the mobile (A) are colinear.

2. Calculate the average speed of A between t_1 and t_6.

$$v_{av} = \frac{d}{\Delta t} \Rightarrow v_{16} = \frac{A_1A_6}{t_6 - t_1} = \frac{A_1A_6}{5\tau} \Rightarrow v_{16} = \frac{10\ cm}{5 \times 50\ ms} = \frac{100\ mm}{250\ ms} = 0.4\ m/s$$

3. Calculate the instantaneous speed of A at t_1, t_3, t_4 and t_6.



Doc. 2

- $v_1 \approx v_{02} = \frac{A_0A_2}{t_2 - t_0} = \frac{A_0A_2}{2\tau} = \frac{15\ mm}{100\ ms} = 0.15\ m/s$
- $v_3 \approx v_{24} = \frac{A_2A_4}{2\tau} = \frac{35\ mm}{10\ \ ms} = 0.35\ m/s$
- $v_4 \approx v_{35} = \frac{A_3A_5}{2\tau} = \frac{45\ mm}{100\ ms} = 0.45\ m/s$
- $v_6 \approx v_{57} = \frac{A_5A_7}{2\tau} = \frac{65\ mm}{100\ ms} = 0.65\ m/s$